

Abstract

Malignant melanoma is an aggressive skin cancer where early detection is paramount for survival. Although Deep Learning has achieved expert-level diagnostic potential [1], deploying these models in portable clinical tools remains a challenge. This project introduces a **Dual-Mode AI-Assisted Dermatoscope** designed to seamlessly bridge clinical examination with automated diagnosis. Building upon advanced videocapillaroscopy architectures [2], we engineered a hybrid optical system featuring a flip mirror mechanism to switch between direct visual observation and digital capture. To enhance diagnostic precision, the design incorporates **Cross-Polarization** and **Green (540nm) Light** for superior vascular contrast. The system integrates a lightweight **EfficientNet-B4** model to classify seven distinct skin lesion types in real-time. Validated on the HAM10000 dataset with **84.63% accuracy**, this device serves as an immediate, automated second opinion, aiming to augment clinical decision-making and accuracy.

Motivation

Early melanoma detection is critical yet often subjective. To address this, we designed a **dual-mode optical system**. While providing high-fidelity data to ensure reliable AI analysis, it simultaneously offers specialized observational modes for direct clinical assessment. This hybrid approach bridges the gap, supporting both automated screening and expert visual validation.

Hybrid Optical Architecture (Dual-Path)

To accommodate both traditional workflow and AI analysis, a custom 4-group optical train was designed.

System Configuration (Fig 1):

- Dual-Path Switching:** A pivotal **Flip Mirror** directs light either to an **Eyepiece** for immediate visual assessment or to a **CMOS Sensor** for digital processing. This modular approach aligns with modern exoscope designs [2].
- Aberration Correction (G1 & G4):** Cemented achromatic doublets minimize chromatic aberration. This ensures high color fidelity, which is critical for identifying subtle pigment patterns in malignant tissues.
- Variator & Compensator (G2 & G3):** Internal movable groups responsible for magnification change (zooming) and focus retention.

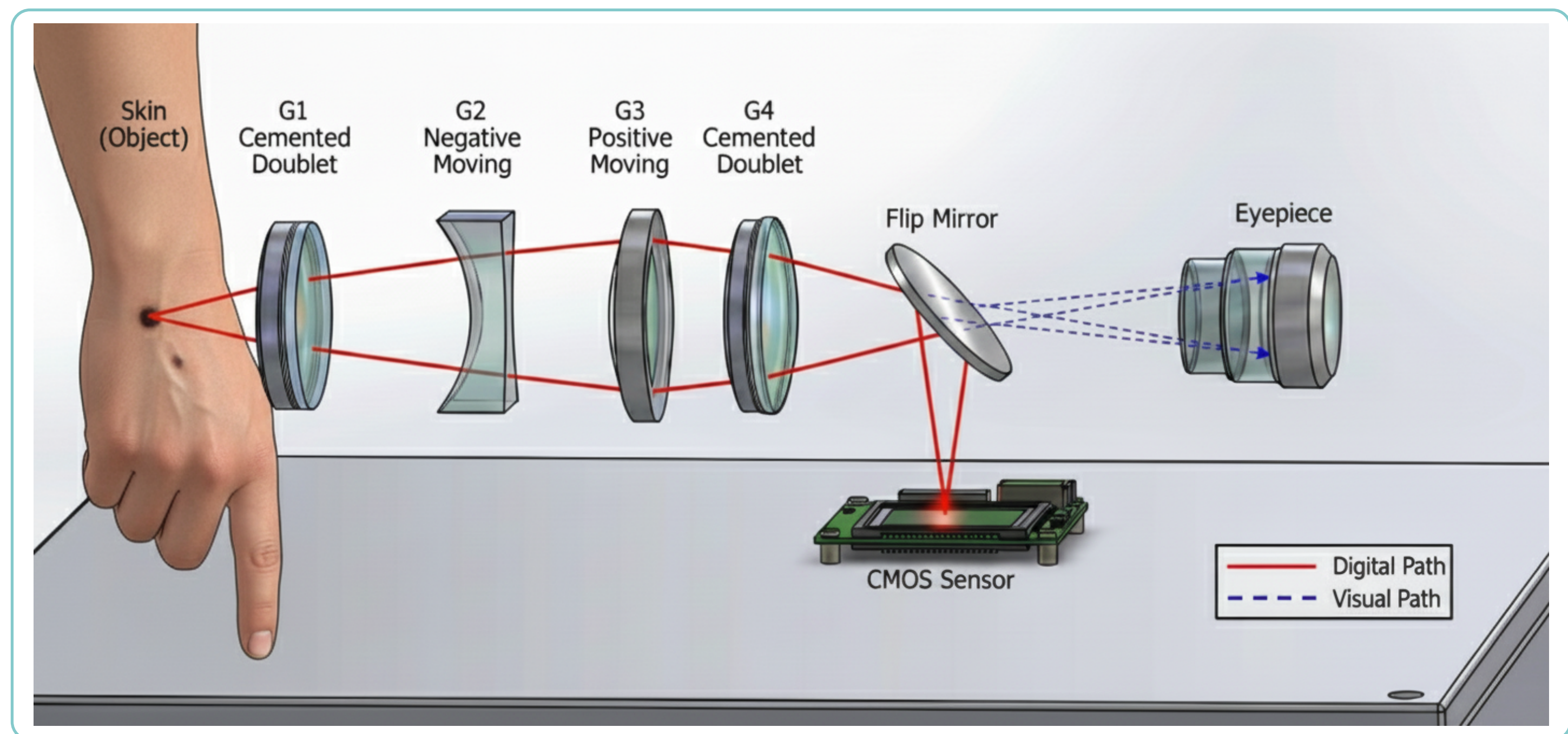


Fig 1. Schematic of the hybrid optical train showing the flip mirror mechanism.

Multispectral & Polarized Illumination

A specialized LED ring delivers cross-polarized and multispectral illumination, ensuring glare-free imaging and enhanced vascular contrast.

A. Cross-Polarization (Glare Removal):

- Technique:** Linear polarizers (0°) on LEDs and an orthogonal analyzer (90°) before the sensor block specular reflection from the stratum corneum, revealing deep tissue structures.

B. Multispectral Imaging (White + Green):

- Broadband White:** Ensures high CRI (Color Rendering Index) for standard pigment analysis.
- Narrowband Green ($\lambda \approx 540\text{nm}$):** Selected to match the **absorption peak of hemoglobin**. This maximizes the contrast of capillary loops and vascular structures, which are critical biomarkers for malignancy [2].

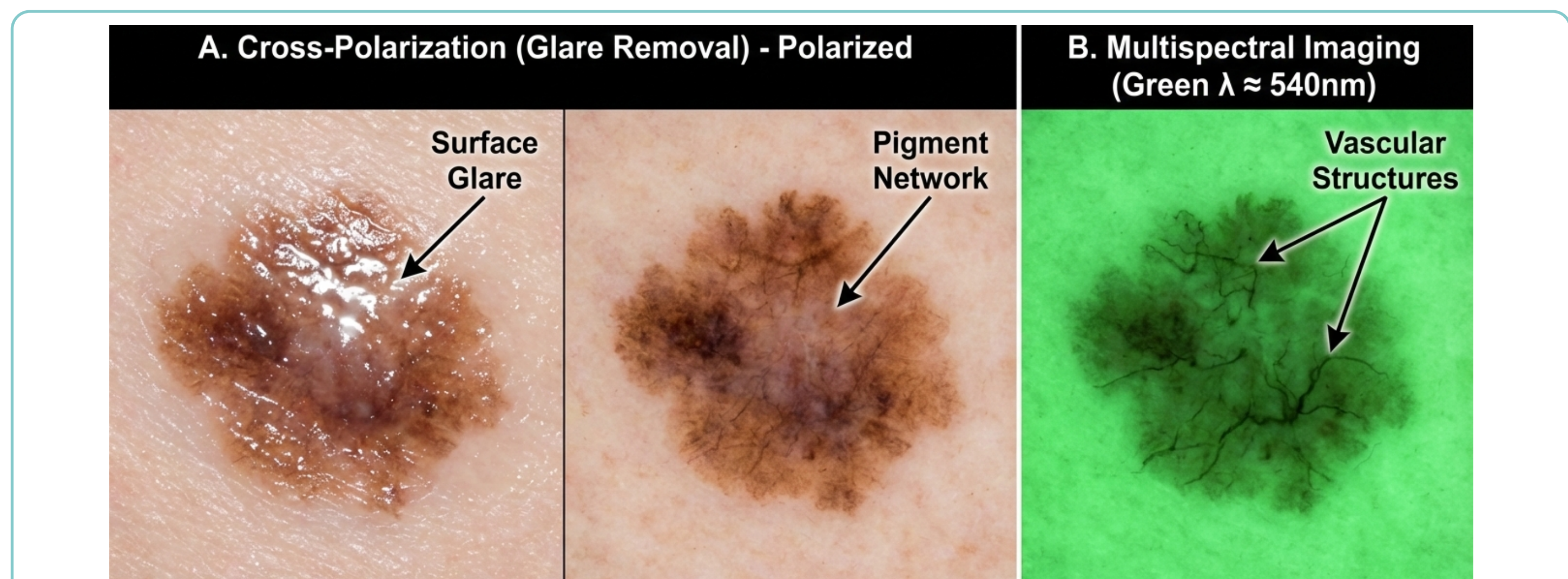


Fig 2. (A) Cross-polarization reveals subsurface pigments by removing glare. (B) Green light ($\lambda \approx 540\text{nm}$) maximizes vascular contrast.

AI Diagnostic Core (EfficientNet-B4)

Digital images are processed by a CNN developed using **PyTorch**.

Dataset: HAM10000 (10,015 images, 7 classes) [3].

- nv (Melanocytic nevi):** Common benign moles.
- mel (Melanoma):** The most aggressive skin cancer.
- bkl (Benign keratosis):** Seborrheic keratoses solar lentigines.
- bcc (Basal cell carcinoma):** Common cancer, low metastasis.
- akiec (Actinic keratoses):** Pre-cancerous lesions.
- vasc (Vascular lesions):** Benign blood vessel lesions.
- df (Dermatofibroma):** Benign skin nodules.

Model Architecture: EfficientNet-B4 Selected for its superior accuracy-to-efficiency ratio via compound scaling, making it suitable for potential embedded use [4].

Handling Imbalance: A **Weighted Random Sampler** was implemented to oversample minority classes (e.g., Melanoma) during training, preventing bias towards benign nevi.

Performance Metrics

In medical image analysis, especially with imbalanced datasets, accuracy alone can be misleading. We prioritized the following metrics to ensure clinical safety:

- Precision:** The ratio of true positive predictions. High precision is essential to minimize *False Positives* (reducing unnecessary biopsies).
- Recall (Sensitivity):** The ability to detect actual positive cases. This is the most critical metric for **Melanoma** to minimize *False Negatives* (preventing missed diagnoses).
- F1-Score:** The harmonic mean of Precision and Recall. It provides a balanced assessment of model performance despite class imbalance.

Experimental Validation

Performance evaluated on an independent test set ($N = 1503$).

Class	Diagnosis	Precision	Recall	F1	Support
akiec	Actinic Keratoses	0.77	0.84	0.80	49
bcc	Basal Cell Carcinoma	0.80	0.86	0.83	77
bkl	Benign Keratosis	0.73	0.72	0.73	165
df	Dermatofibroma	0.88	0.88	0.88	17
mel	Melanoma (Malignant)	0.57	0.71	0.63	167
nv	Melanocytic Nevi	0.94	0.89	0.91	1006
vasc	Vascular Lesions	0.83	0.91	0.87	22
AVG	Weighted Average	0.86	0.85	0.85	1503

Overall Accuracy: 84.63%

Clinical Interpretation:

- Melanoma Sensitivity:** The model achieved a **Recall of 0.71** for Melanoma. The trade-off with Precision (0.57) prioritizes a safety-first approach (minimizing false negatives).
- System Synergy:** The multispectral, cross-polarized optical front-end enhances feature extraction for the AI model [5].

Conclusion & References

This project validates a **Dual-Mode System** that seamlessly merges the objectivity of Deep Learning with the versatility of optical engineering. The embedded **EfficientNet-B4** model serves as a critical "second opinion," capable of classifying seven distinct lesion types with a robust **84.63% accuracy**. By analyzing glare-free, cross-polarized images provided by the optical front-end, the AI mitigates human subjectivity and reduces the risk of overlooking malignant cases. Simultaneously, the system preserves the clinician's role through its versatile hardware. The flip-mirror and Green Light (540nm) modes allow for immediate, detailed manual verification of the AI's predictions. Ultimately, this prototype offers a complete diagnostic workflow: reliable, automated screening for **early detection**, coupled with specialized optical tools for expert clinical validation.

References:

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